

10:05

5G 21



0an0stn3n3q.archer.replit.dev



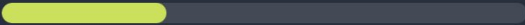
>_

No comparison yet

Every result must round-trip before it can be selected.

The first-run benchmark

Compression is data-dependent. On a representative 1,024 byte input, the lab uses 320 bytes as an orientation target — never as a promise.

1,024 B  320 B

31.25% of original · 704 B saved

What “verified” means

The runtime compresses locally, then checks that the output can be decoded back to the exact original bytes. Zstandard runs at level 22; PAQ 9 and the selected Fast or Ultra transform search are browser-local and fully round-trip tested.

 The saved .clz file stores its codec, transform parameters, and original filename for safe extraction.



10:05

5G 21



0an0stn3n3q.archer.replit.dev



Fast · 256 base transforms



TIME LIMIT (SECONDS)

10

Enter 10 for a ten-second search. Clear the field to run until the selected transform set is complete.

Candidate outputs

Load bytes to populate a verified comparison.

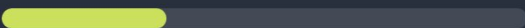
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What "verified" means



The runtime compresses locally, then checks that the



PAQJP 8.5



PRECISION DATA COMPRESSION
WORKBENCH

ENGINE STATUS

● **ONLINE**



TRANSFORMS

256



PAIR SEQUENCES

2704



BACKENDS

PAQ

ZSTD-22

RAW



COMPRESS



EXTRACT



CHECK



Made with Replit ×



The screenshot displays a mobile IDE interface with a sidebar on the left containing icons for file explorer, search, Git, testing, workspace, terminal, and user profile. The main area is divided into three panes: a file explorer, a code editor, and a terminal.

File Explorer: Shows the project structure for 'ADVANTAGE2 [CODESP...]'.

- ADVANTAGE2 [CODESP...]
 - > .circleci
 - > .devcontainer
 - > assets
 - > Dictionaries
 - > src
 - > static
 - > tests
 - > .gitignore
 - 1.py (1, U)
 - app.py
 - CONTRIBUTING.md
 - demo_callbacks.py
 - demo_configs.py
 - demo_interface.py
 - LICENSE
 - README.md
 - requirements.txt

Code Editor: Displays the code for '1.py'.

```
3172 def main():
3240
3241
3242
3243 if __name__
3244     main()
```

Terminal: Shows the output of running the code.

```
aces/advantage2 $
cd
/workspaces/
advantage2 ; /usr
...
... 50000 indic
es tested OK
... 60000 indic
es tested OK
All 65536 trans
formations are lo
ssless on test by
te.

Random 1000-byte
pipeline test (LZ
H+RLE+backend)...
PASS

[All checks passe
d - 100% lossless
]
```

Status Bar: Shows 'Codespaces: miniature', 'main*', and various icons.



10:05

5G 21



0an0stn3n3q.archer.replit.dev



Or paste bytes

DIRECT INPUT

UTF-8 text

Hex pairs

Paste a string to inspect...



stays in this tab

Load bytes

Transform search

OPTIONAL PRECONDITIONING

MODE

Fast · 256 base transforms



TIME LIMIT (SECONDS)

10

Enter 10 for a ten-second search. Clear the field to run until the selected transform set is complete.

Candidate outputs

Load bytes to populate a verified comparison.



No comparison yet

Every result must round-trip before it can be

w57w9g4p7v39jr.github.dev

E Energia |...

DCU to the si...

8

E Energia |...

E Energia |...

My Even...

welcom...

advantage2 [Codespaces: jubilant ...]

SOURCE CONTROL

CHANGES

Message (Æ...)

Commit

Changes 1

1.py

Auto

Merg...

main

Fix double semico...

TERMINAL

Python Debug Console

advantage2 ; /usr/bin/env /usr/local/bin/python /home/user/.vscode-remote/extensions/ms-python.debugpy-2026.6.0-linux-x64/bundled/libs/debugpy/adapter/../../debugpy/launcher 36707 -- /workspaces/advantage2/1.p

...

Testing random 1000-byte block through full compress/decompress...

PASS: random data pipeline OK in both modes

Testing empty input...

PASS: empty input pipeline OK

[All tests passed - compressor is 100% lossless]

ocean-dev /workspaces/advantage2 \$

10:05

5G 22



0an0stn3n3q.archer.replit.dev



Find the smallest verified output.

Compare Zstandard, PAQ, and a reversible transform search. Every result is decoded and checked before it can win.

▶ Run comparison



Load a file from this device

Bytes are read into memory only. No upload endpoint, no account, no history.

Choose file ↓

Or paste bytes

DIRECT INPUT

UTF-8 text

Hex pairs

Paste a string to inspect...



15:31

5G 33



TRANSFORMS

256



PAIR SEQUENCES

2704



BACKENDS

PAQ

ZSTD-22

RAW



COMPRESS



EXTRACT



CHECK



Click or drag
file to this area
to upload

Select a file to
compress losslessly



Made with Replit ×

3f-b52c097e0134-00-1sp7g069icftj.picard.repl.co — Private

advantage2 [Codes...]

- > .circleci
- > .devcontainer
- > assets
- > Dictionaries
- > src
- > static
- > tests
- > .gitignore
- 1.py 1, U
- app.py
- CONTRIBUTING.md
- demo_callbacks.py
- demo_configs.py
- demo_interface.py
- LICENSE
- README.md
- requirements.txt

aces/advantage2 \$

```

sr/bin/env /u
sr/local/bin/pyth
...
to be skipped.
NOTE: this is a
sanity check only, not a proof of
universal losslessness.
Actual compression verifies every
candidate transform on the real
input;
unsafe transforms are rejected,
and raw fallback guarantees lossless
output.

Random 1000-byte
pipeline test (LZ
H backend)...
PASS

[All checks passed - 100% lossless]

Menu:
1) Compress (Fast) - only 256 single transforms
2) Compress (Ultra)

```

Codespaces: miniature main* 0 1 2

rcwqpp.github.dev



≡ 5xgrx6w5v99frp7.github.dev ↻



UPGRA...



UPGRA...



The Lea...



Collectio...



Sign in t...

Sign in t...



welcom...



advantage2 [Codespaces: musical...]



SOURCERY



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1

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TERMINAL ...



Python Debug Console ⚠️



```
dvantage2 ; /usr/bin/env /usr/local/bin/python /home
/user/.vscode-remote/extensions/ms-python.debugpy-20
26.6.0-linux-x64/bundled/libs/debugpy/adapter/../../
debugpy/launcher 39843 -- /workspaces/advantage2/1.p
y
```

Testing empty input...

PASS: empty input pipeline OK

[All tests passed – compressor is 100% lossless]

⊗ ocean-dev /workspaces/advantage2 \$ 3

bash: 3: command not found

⊗ ocean-dev /workspaces/advantage2 \$ 3

bash: 3: command not found

○ ocean-dev /workspaces/advantage2 \$ █

15:26

5G 33



Agent



Worked for 11 minutes



Please add menu option 3 check of 2704
trnfomations

4 hours ago



(10 seconds)



Pasted--usr-bin...

Please full app with compress and extract put
with 2704 transformations lossless always 100%
after paq level 9 and zstandard level 22 the same
how in the code without bugs and errors Option 3
check Please add menu option 3 check of 2704
trnfomations

↓ Scroll to latest

4 hours ago

Make, test, iterate...



Plan

Economy ▾



replit.com



Source | advantage2... | Console

Repositories

- adv...

Changes (3)

- 1... 2, U
- Myself.docx U
- Myself.docx.a1 U

Commits

- Merg... main
- Add comments to...
- Add PR template ...
- Replace dwave_n...
- Merge pull reque...
- Remove minor ve...
- Merge pull reque...
- Fix double semic...
- Merge pull reque...
- Disable run butto...
- Merge pull reque...
- Refactor qpu loo...
- Update fonts Kat...
- Actually update d...
- Run isort and ble...

```

antage2 $ cd /wor
/advantage2 ; /usr
v /usr/local/bin/p
home/user/.vscode-
...
Qubits=16 Pairs=1
ide=3
Transforms 1..25
ngle bytes: PASS
Transforms 1..25
lti-byte vectors:
Testing user qub
sform (ID 60) ...
User qubit trans
6q): PASS
Testing 8192-qub
it transform (ID 6
8192-qubit Qiski
form: PASS
Testing headerle
le-Diameter-Dot tr
(ID 61) ...
Headerless Circl
ter-Dot transform:
Testing XML/DOCX
ansform (ID 63) ..
XML/DOCX/DOC tra
PASS
Testing Lepton J
nd-trip (if availa
.
Lepton JPEG roun
PASS (633 -> 527
Testing minus tr
s (sampled) ...
Minus transforms
ed): PASS
First 1000 pairs
Backends: PASS
[ALL LOSSLESS CHEC

```



vh69gv.github.dev



The screenshot displays a mobile web interface for a code editor. At the top, the status bar shows the time 2:46, 5G connectivity, and a battery level of 39%. The main interface is divided into three sections:

- Left Sidebar:** Contains navigation icons for Explorer, Search, Source Control (13), Run and Debug (1), Extensions, Testing, GitHub, and Python.
- Explorer Panel:** Shows the file structure of the 'advantage2' project. The files listed are:
 - .circleci
 - .devcontainer
 - assets
 - Dictionaries
 - src
 - static
 - tests
 - .gitignore
 - 1.py (1, U)
 - app.py
 - CONTRIBUTING.md
 - demo_callbacks.py
 - demo_configs.py
 - demo_interface.py
 - LICENSE
 - README.md
 - requirements.txt
- Editor Panel:** Displays the content of the selected file, '1.py'. The text includes:


```

aces/advantage2 $
sr/bin/env /u
sr/local/bin/pyth
...
es tested OK
... 40000 indic
es tested OK
... 50000 indic
es tested OK
... 60000 indic
es tested OK

All sequence in
dices processed o
n test bytes 0x00
, 0xFF, 0xAA, 0x5
5.

Mismatches woul
d have caused fai
lure; TransformEr
rors are allowed
to be skipped.

NOTE: this is a
sanity check onl
y, not a proof of
universal lossle
ssness.

Actual compress
ion verifies ever
y candidate trans
form on the real
input;
unsafe transfor
ms are rejected,
and raw fallback

```

At the bottom, a blue status bar indicates 'Codespaces: miniature', 'main*', and various icons for file operations and notifications.

